

# **Opportunity, Affordance, and Action Availability**

*Toward a Formal Account of What an Agent Can Perceive, Believe, and Do*

Cross-disciplinary research paper

# Abstract

Action availability is represented differently across ecological psychology, human–computer interaction, decision theory, control, reinforcement learning, planning, formal methods, and computer security. These traditions distinguish real possibilities for action, actions enabled in a state, authorized operations, reachable outcomes, perceptually indicated operations, believed possibilities, selected acts, attempted executions, successful transitions, and recovery. No single established construct captures all of these dimensions without changing its inherited meaning. Ecological affordance is relational but does not by itself encode authority, formal state enablement, interface exposure, explicit signaling, belief, execution success, or reversibility. Action-space and choice-set formalisms precisely encode alternatives but usually abstract away their epistemic presentation. Authorization models separate authority from technical capability, while transition systems and control theory distinguish enabled actions and reachable states. HCI separates physical or functional possibilities from signifiers, discoverability, and learned interpretation.

This paper concludes that a cross-disciplinary opportunity model is justified, but opportunity is not a new primitive. It is an analytical coordination role over established typed relations. The proposed model represents each candidate action by system-relative feasibility, permission, state enablement, reachability, projection exposure, signaling, agent-relative belief, expected consequences, selection, execution, success, reversibility, and uncertainty. The framework is formalizable as linked predicates over a transition system, authorization relation, projection mapping, observation model, and agent belief state. It provides comparison rules for semantically equivalent projections without implying behavioral equivalence. Six hypotheses are operationalized, and reusable vocabulary, matrices, contracts, comparison protocols, and evidence maps are supplied.

## 1. Problem Statement

Computational environments frequently support actions that are hidden, display controls that are currently disabled, imply operations they cannot perform, or expose equivalent operations differently across modalities. Treating all such cases as “affordances” obscures important distinctions. A technically valid operation may be prohibited by policy. A permitted operation may not be enabled in the current state. An enabled operation may be absent from the current projection. An exposed operation may not be signaled. A signal may be perceived but not interpreted as actionable. An attempted operation may fail, and a successful transition may still fail to achieve the agent’s intended outcome.

The explanatory problem is therefore not merely to enumerate actions. It is to state which action relation is being claimed, relative to which system state, authority, projection, agent, task, and time. The central question is how action opportunities should be represented, measured, and compared across projections while preserving distinctions among system-relative availability, epistemic access, belief, choice, execution, outcome, and recovery.

## 2. Literature Review Method

The review used a concept-centered comparative method. Primary and foundational sources were prioritized where available: Gibson’s ecological account of affordance; Norman’s distinction between affordances and signifiers; Gaver’s technology-affordance analysis; formal transition-system notions of enabled action and reachability; Markov decision-process and options formalisms; control-theoretic controllability; planning operators and preconditions; access-control models; and empirical HCI work on discoverability, feedback, and error recovery. Secondary sources were used only to locate terminology or identify disputes.

The analysis proceeded in four stages. First, each tradition’s explanandum and formal commitments were identified. Second, candidate opportunity distinctions were tested for logical independence using counterexamples. Third, the distinctions were mapped to existing terminology to avoid unnecessary invention. Fourth, the resulting model was evaluated against six falsification conditions: theoretical redundancy, measurement failure, lack of predictive gain, disappearance of projection effects under controls, excessive context dependence, and existence of a single complete formalism.

## 3. Affordance Traditions

Gibson introduced affordance as what an environment offers an animal, emphasizing complementarity between animal and environment rather than a property located exclusively in either. On this account, an affordance can exist whether or not it is currently noticed, because it is grounded in the relation between environmental structure and the actor’s effectivities. Perception concerns information specifying that relation. Successful action is not required for the affordance to exist, although empirical affordance research often evaluates perceptual calibration by comparing judgments with action limits.

HCI adopted and transformed the term. Norman initially used perceived affordance to emphasize what users believe can be done, then later separated affordances from signifiers: the action possibility is one matter; the perceptible indication of where or how to act is another. Gaver distinguished perceptible, hidden, false, and correct affordances, revealing a two-dimensional space between actual action possibility and perceptual information. Hartson further divided physical, cognitive, sensory, and functional affordances,

but this expands “affordance” toward a taxonomy of interaction support rather than preserving Gibson’s narrower ecological relation.

The literature therefore contains several non-equivalent affordance traditions:

- Ecological affordance: a relation between actor effectivities and environmental properties that makes an action possible.
- Perceived affordance: an agent’s perception or belief that an action is possible.
- Signifier: perceivable evidence indicating an operation, location, or manner of action.
- Technological or functional affordance: a designed system’s support for a class of action or use.
- Social affordance: possibilities for social action structured by other agents, conventions, or institutions.
- Learned or cultural affordance: action possibilities whose detection or uptake depends on acquired practices.

These uses should not be collapsed. Ecological affordance is strongest for relational physical possibility; signifier is strongest for perceptual indication; perceived affordance and learned affordance address agent-relative interpretation; functional affordance often approximates system capability. None alone supplies authority, formal reachability, execution success, or recovery.

## 4. Action-Set and Choice-Set Traditions

Decision theory and economics represent alternatives through feasible sets or choice sets. These specify options available for selection under constraints. Game theory uses action sets or strategy sets, often indexed by player and information state. Reinforcement learning uses an action set  $A$ , commonly with a state-dependent subset  $A(s)$ , and a policy mapping states or beliefs to action distributions. The options framework adds temporally extended actions with initiation sets, internal policies, and termination conditions. Planning formalisms represent operators with preconditions and effects. Formal transition systems define an action as enabled in a state when a corresponding transition exists. Control theory distinguishes admissible controls and reachable states. Robotics often uses affordance models to prune actions by predicting whether an action is feasible or useful in a state.

These formalisms provide exact answers to different questions. A state-dependent action set identifies primitive alternatives available to a model. A choice set describes alternatives offered to a decision-maker. An enabled action is one whose preconditions hold. Reachability concerns whether a target state can be attained by a path, not whether a single action is currently executable. An option is a temporally extended policy, not simply any opportunity. A controllable input is an admissible intervention under a system model. A planning operator is a symbolic transition schema. These concepts are compatible but not equivalent.

## 5. Capability, Permission, and Reachability

Capability, permission, and reachability must remain separate.

Capability has two established senses. In ordinary agent modeling, it denotes an agent’s ability or resources to perform an action. In capability-based security, it is an unforgeable token of authority naming an object and associated rights. The latter is not equivalent to physical or computational ability. To avoid ambiguity, the present framework uses effective capability for the conjunction of required agent and system capacities, and authority capability only when referring to security tokens.

Permission is normative or institutional. Access-control systems define whether a subject is authorized to perform an operation on an object under a policy, role, attributes, and context. A system may be technically able to execute a prohibited action; prohibition does not make the transition physically or computationally impossible.

State enablement is local. An action is enabled when its preconditions hold in the current state. Reachability is path-relative. An outcome is reachable if there exists an admissible sequence from the current state to that outcome. An action can be permitted yet disabled, enabled yet lead nowhere useful, or currently disabled while its outcome remains reachable through an alternative sequence.

This yields four system-relative relations:

1. Domain-valid: the operation belongs to the declared domain model.
2. Effective-capable: the agent-system coupling has the capacities required for execution.
3. Permitted: policy authorizes the operation.
4. Enabled: the operation’s current-state preconditions hold.

Reachability is then defined over sequences of enabled and permitted transitions, with any safety, resource, and temporal constraints stated explicitly.

## 6. Discoverability, Signaling, and Belief

Exposure, signaling, discoverability, and belief describe different epistemic relations.

Exposure means that the current projection provides an executable access path to an operation. A menu item, command syntax, API endpoint, gesture, or conversational intent can expose an operation. Exposure does not require conspicuous indication: a keyboard shortcut may be exposed but hidden from a novice.

Signaling means that perceivable structure indicates an operation, its locus, or its expected manner of use. Labels, shape, contrast, prompts, examples, disabled states, documentation, and feedback can signal. A signal can be false when it suggests an unavailable operation.

Discoverability is an empirical relation among projection, agent, task, context, and search process. It concerns whether and how an agent can find an operation or relation without already knowing the access path. Discoverability should be measured through controlled search, recognition, recall, or guided exploration—not inferred from visibility alone.

Believed action is an action the agent represents as available with some confidence and expected consequence. Belief is latent. It requires elicitation, prediction, confidence reports, recognition tests, or model-based inference with acknowledged uncertainty. Selection is observable choice among represented alternatives. Execution is an observable attempt. Success is a system-relative transition criterion or task-relative outcome criterion; these must be distinguished.

## 7. Candidate Typed Opportunity Model

The evidence supports a typed opportunity model, not opportunity as a primitive natural property. “Opportunity” is retained as an umbrella analytical role because no single inherited construct coordinates domain validity, capability, authority, state enablement, reachability, exposure, signaling, belief, outcome expectation, and recovery without semantic overextension.

Let a computational interaction model be:

$$M = \langle S, A, T, C, P, \Pi, \Sigma, O, B, G, R \rangle$$

where  $S$  is the set of system states;  $A$  is the action vocabulary;  $T \subseteq S \times A \times S$  is the transition relation or probabilistic kernel;  $C(a, x, s)$  is effective capability for agent  $x$ ;  $P(a, x, s)$  is permission;  $\Pi$  maps system state and action structure into a projection;  $\Sigma$  is the signaling relation realized by the projection;  $O$  is the observation model available to the agent;  $B_x$  is the agent's belief distribution over states, actions, and consequences;  $G$  is the task or goal specification; and  $R$  describes recovery and reversibility relations.

For agent  $x$  at time  $t$  in state  $s$  under projection  $\pi$ , the opportunity profile is not one set but a family of typed sets and relations:

$D(s)$ : domain-valid actions.

$C(s)$ : effectively capable actions.

$P(s)$ : permitted actions.

$E(s)$ : enabled actions.

$X\pi(s)$ : exposed actions.

$S\pi, x(s)$ : signaled actions detectable by  $x$ .

$B(s)$ : actions believed available by  $x$ , with confidence.

$K(s, a)$ : expected consequence distribution.

$L(s, G)$ : reachable outcome set under admissible sequences.

$V(s, a)$ : reversibility or recoverability relation.

$U$ : uncertainty attached to system state, observer state, and inferred variables.

Selected, executed, and successful are event predicates over a trace rather than standing opportunity sets. This avoids treating behavior as part of availability.

The currently executable authorized set is:

$$A_{\text{exec}}(x, s) = D(s) \cap C(s) \cap P(s) \cap E(s).$$

The projection-accessible set is:

$$A_{\text{access}}(x, s, \pi) = A_{\text{exec}}(x, s) \cap X\pi(s).$$

The signaled executable set is:

$$A_{\text{signal}}(x, s, \pi) = A_{\text{access}}(x, s, \pi) \cap S\pi, x(s).$$

The believed executable set is not assumed to be a subset of  $A_{\text{exec}}$ :

$$A_{\text{belief}}(x, s, \pi) = B(s).$$

False positive belief consists of  $B(s) \setminus A_{\text{exec}}(x, s)$ ; hidden opportunity consists of  $A_{\text{exec}}(x, s) \setminus B(s)$ . This asymmetry is central to conversational overclaim, expert knowledge, and misleading controls.

## 8. Formal Representations

No single formalism covers every distinction without extension.

Labeled transition systems are best for state, enabled action, trace, and reachability. Planning domains add preconditions, effects, and goals. MDPs and POMDPs add uncertainty, value, policies, and belief states. Options add temporally extended action. Control models formalize admissible inputs, dynamics, and controllability. Access-control matrices and policy languages formalize authorization. Observation models formalize partial access to state. HCI and ecological models contribute signaling, perception, effectivities, and learned interpretation. Reversibility requires inverse transitions, compensating actions, checkpoints, or probabilistic recoverability.

The recommended architecture is layered but linked:

Domain and transition layer: D, S, A, T.

Capability and authority layer: C, P.

Projection layer:  $\Pi$ , exposure X, signaling  $\Sigma$ .

Epistemic layer: O, belief B, expected consequences K.

Policy layer: goals G and action-selection policy  $\mu$ .

Trace layer: selection, execution, transition, outcome, feedback.

Recovery layer: V and repair paths.

The layers should not be treated as a one-way causal chain. Projection can alter actual feasibility when it gates operations, not merely epistemic access. Execution changes state and therefore subsequent opportunity. Feedback changes beliefs, skill, and policies. Authority can change dynamically. Social action can alter policy or projection. The model is therefore a coupled state-transition system with feedback.

## 9. Empirical Measurement

System-relative variables can often be observed or computed directly: domain membership, permission decision, precondition satisfaction, transition occurrence, reachable states under a declared model, exposure path, and recorded execution. Signaling strength, belief, expected consequence, confidence, and perceived reversibility require empirical measurement or inference.

Recommended measures include:

Action discovery: proportion finding an operation; time to first correct identification; search path length; number of representation regions inspected; unaided versus prompted discovery.

Awareness and belief: recognition of available actions; free recall; prediction of whether an operation will execute; confidence-calibrated availability judgments; precision and recall of the believed action set against A\_exec.

Selection: conditional probability of selection given belief and exposure; choice latency; switching rate; strategy entropy.

Unavailable-action attempts: frequency and type of actions executed outside E, P, or D; whether the mismatch was signaled; recovery after rejection.

Recovery: time to identify corrective action; repair success; number of irreversible or escalating transitions; use of undo, backtracking, help, or alternative routes.

Transfer: change in believed action precision, discovery time, and strategy use across projections or tasks, controlling for exposure history.

Reversibility judgment: predicted recoverability, confidence, and calibration against the actual recovery graph.

Direct observations should be separated from latent interpretations. A pause is not direct evidence of hesitation; it is a duration requiring alternative explanations. An unselected action is not evidence of unawareness. Task completion is not evidence of equal opportunity. Belief claims require either explicit reports or a validated latent-variable model.

## 10. Comparative Cases

Case 1 — Hidden but feasible action. A system supports an API operation that is permitted, enabled, and executable, while the GUI omits it. The action is in A\_exec but not X\_GUI. For an expert using documentation it may be signaled and believed through another projection; for a GUI-only novice it is hidden. Projection changes epistemic and access structure, not the underlying transition relation.

Case 2 — Visible but unavailable action. A disabled control is exposed as a representational object and may signal a future or conditional operation, but it is not currently executable. It is not a current action opportunity under  $A_{\text{exec}}$ . It is a signifier of a domain-valid action whose enablement condition fails. This can improve learnability if the condition is interpretable, or create false belief if the reason is opaque.

Case 3 — Conversational overclaim. A dialogue system states that it can delete an account although no authorized backend operation exists. The action is signaled and may enter the user's believed set but is absent from D, P, E, or X at the execution layer. This is a false-positive opportunity claim. The model predicts elevated failed attempts, trust calibration errors, and inappropriate planning.

Case 4 — Accessibility substitution. A single operation is exposed visually, auditorily, and tactually. Domain validity, permission, and transition semantics may remain invariant, while signaling latency, sequence, spatial comparability, memory demand, error detectability, and recovery differ. Equal task completion does not establish equivalent discovery or effort. Modality-specific observer assumptions must be declared.

Case 5 — Expert and novice. The projection exposes the same executable operations. The system-relative sets D, P, E, and X are equal. Signaling detectability and believed action sets differ because conventions, vocabulary, and procedural knowledge differ. Training can change B and policy  $\mu$  without changing T or X.

Case 6 — Policy and authorization. A data-export operation is semantically valid, technically executable, and visible, but the user's role is denied. Capability and exposure are present; permission is absent. Attempts may be blocked before transition. The correct explanation is authorization, not lack of affordance or reachability.

## 11. Operationalized Hypotheses

### H1 — Signaling effect

Exposure: manipulation of signifier presence, specificity, or state indication while executable action sets remain fixed. Outcome: unaided discovery rate, discovery time, and conditional selection. Controls: layout, motor distance, latency, task value, familiarity, operation order. Manipulation check: recognition of the cue and its intended referent. Exclusions: prior knowledge of hidden shortcut where it defeats manipulation; inaccessible modality. Analysis: multilevel logistic model for discovery and survival or accelerated-failure-time model for latency. Falsification: no practically meaningful signaling effect with confidence interval excluding the preregistered minimum.

### H2 — Belief mediation

Exposure: projection exposure and signaling condition. Mediator: preregistered availability-belief score measured before selection. Outcome: selection and correct execution. Controls: task value, skill, motor cost, confidence response bias, prior exposure. Manipulation check: exposure and signal detection. Exclusions: belief measured after outcome feedback. Analysis: causal mediation with measurement-error model and temporal ordering. Falsification: exposure affects selection without a reproducible indirect effect through measured belief, or belief lacks predictive validity.

### H3 — Opportunity inequivalence

Exposure: two semantically equivalent projections with identical operation semantics and task outcomes. Outcome: typed opportunity distance across exposure, signaling, belief precision/recall, path cost, recovery visibility, and auditability. Controls: state, authority, device performance, content, task, training. Manipulation check: semantic and reachable-outcome equivalence. Exclusions: projections that change backend permissions or transitions unintentionally. Analysis: equivalence/non-equivalence tests by dimension. Falsification: all preregistered dimensions meet equivalence bounds.

### H4 — Training effect

Exposure: structured training versus control under a fixed projection. Outcomes: believed-action precision/recall, discovery time, strategy diversity, and transfer. Moderator: prior expertise. Controls: repetition, feedback, task difficulty, motivation. Manipulation check: knowledge test without execution cues. Exclusions: system or projection changes during study. Analysis: mixed-effects longitudinal model. Falsification: training produces no durable or transferable change beyond repetition.

### H5 — Recovery effect

Exposure: recovery actions exposed/signaled versus hidden or ambiguous, crossed with actual reversibility. Outcomes: error detection, repair initiation, successful recovery, escalation, perceived safety. Mediators: believed recovery set and perceived reversibility. Controls: error severity, time pressure, prior error experience, feedback latency. Manipulation check: recognition of

recovery cue. Exclusions: failures with external irreversible consequences not represented in the model. Analysis: multilevel event-history and mediation models. Falsification: recovery support does not alter repair behavior or measured belief under adequate power.

## **H6 — Opportunity model gain**

Exposure: observational or experimental dataset spanning projections, states, permissions, and expertise. Outcome: held-out prediction of discovery, selection, execution failure, success, and recovery. Predictors: baseline state, task, projection label, history, and action logs versus baseline plus typed opportunity variables. Controls: leakage, model capacity, participant clustering, temporal drift. Manipulation check: reliability of each typed variable. Exclusions: variables derived from future outcomes. Analysis: nested out-of-sample comparison using log loss, calibration, and preregistered minimum gain. Falsification: no robust incremental predictive or explanatory gain.

## **12. Falsification Tests**

The framework should be narrowed or rejected under the following evidence.

1. Affordance sufficiency. If a stable, operational ecological-affordance formalism can encode authority, state enablement, exposure, signaling, agent belief, expected consequence, execution success, and reversibility without redefining affordance beyond recognition, the umbrella model is redundant. Current evidence does not show this.
2. Action-space sufficiency. If state-dependent action sets plus observation models and policies explain projection and belief effects without loss, the opportunity vocabulary can be reduced to those components. This remains a serious alternative. The proposed model therefore claims organizational and comparative value, not a new causal substance.
3. Measurement failure. If believed and signaled sets cannot be estimated with acceptable reliability or predictive validity, they should not be treated as stable measured constructs. They may remain trial-specific latent variables.
4. No incremental gain. H6 is falsified if typed variables do not improve out-of-sample prediction or explanation over state, task, projection label, prior experience, and action logs.
5. Projection null. H1 and H3 are weakened if signaling and exposure manipulations have no effect after familiarity, task, motor cost, latency, and implementation quality are controlled.
6. Context non-comparability. The comparison protocol fails if opportunity differences cannot be stated under common target, task, observer, and tolerance conditions. In that case comparison must remain local rather than general.
7. Formal unification. If one established formalism already supports every required relation with standard terminology and empirical mappings, the cross-disciplinary synthesis should adopt that formalism rather than retain “opportunity.”

## **13. Threats to Validity**

Conceptual breadth is the principal threat. Terms such as affordance, capability, option, and information have discipline-specific meanings, and normalization can erase productive differences. The paper therefore retains typed definitions rather than declaring terminological identity.

Formal models simplify human action. Belief sets, goals, and expected consequences may be distributed, unstable, socially negotiated, or non-propositional. Treating them as variables is an analytical commitment, not a claim that cognition literally stores enumerated sets.

Reachability and reversibility depend on model completeness. Hidden external state, asynchronous effects, institutional consequences, and irreversible social outcomes can make computed recovery misleading.

Empirical identification is difficult. Exposure, signaling, motor cost, familiarity, and implementation quality often covary. Mediation analysis of belief is vulnerable to measurement error and temporal ordering. Demographic categories should not substitute for measured language, experience, perceptual access, training, or situated constraints.

The framework is strongest for declared computational tasks with identifiable operations and state transitions. It is weaker for open-ended creative, social, or interpretive activities in which action vocabularies and goals emerge during activity.

## **14. Implications for Interaction Research**

The model supplies the missing bridge between representation and behavior without claiming that representation determines action. Projection contributes exposure and signals; observation and prior experience support interpretation; interpretation updates

beliefs and expected consequences; policy selects among believed alternatives; execution is filtered by capability, permission, and enablement; outcomes and feedback alter later state, belief, and skill.

Opportunity therefore functions primarily as a structured descriptive and mediating construct. It describes the typed relations that condition action selection. Some components can mediate projection effects—particularly belief and expected consequences—while others moderate them, including permission, capability, task, and expertise. The umbrella term itself is not a causal variable unless operationalized through its components.

For cross-projection comparison, semantic equivalence should be established independently. Two projections can preserve the same operation semantics and reachable task outcomes while differing in exposure, signaling, belief calibration, recovery visibility, sequence cost, and auditability. Such projections are semantically equivalent under one contract but opportunity-inequivalent under another.

## 15. Conclusion

The evidence supports determination 3: a cross-disciplinary opportunity model is justified, but opportunity is not a new primitive.

Existing traditions already supply the constituent concepts. Ecological psychology supplies relational action possibility; HCI supplies signifiers, discoverability, and perceived availability; formal methods and planning supply enabled actions and reachability; control supplies admissible interventions and controllability; decision theory and reinforcement learning supply action sets, beliefs, policies, options, and expected consequences; access control supplies permission and authority; error-recovery research supplies reversibility and repair. None is individually complete for comparing what computational projections make feasible, accessible, apparent, believed, executable, and recoverable.

The warranted contribution is therefore a typed taxonomy and integration contract. Opportunity should denote the coordinated profile of established relations under declared system, projection, agent, task, and temporal conditions. It should not replace affordance, action set, permission, reachability, signifier, belief, policy, or reversibility. Its scientific value depends on measurable components, explicit uncertainty, comparative prediction, and demonstrated incremental explanatory gain.

## 16. Reusable Artifacts

### A. Canonical Vocabulary

| Term               | Canonical definition                                                                                                                                                   | Boundary                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Affordance         | A relation between environmental properties and an agent's effectivities that makes a class of action possible; perception is not constitutive in the Gibsonian sense. | Agent–environment relative; not identical to signal, belief, permission, or success.    |
| Action possibility | A minimally specified relation under which an action can occur given relevant capacities and conditions.                                                               | Use only when the relevant modality—physical, computational, institutional—is declared. |
| Capability         | The effective capacity or resources required for an agent-system coupling to execute an action.                                                                        | Do not confuse with permission or with a security capability token.                     |
| Permission         | Authorization under a policy, role, rule, or institutional condition for a subject to perform an operation.                                                            | Normative; an impermissible action may remain technically possible.                     |
| Reachability       | Existence of an admissible sequence of transitions from a current state to a target state or set.                                                                      | Path-relative, constraint-relative, and model-relative.                                 |
| Exposure           | Provision by the current projection of an executable access path to an operation.                                                                                      | An exposed operation may be hidden, poorly signaled, or unknown.                        |
| Signaling          | Perceivable indication of an operation, its locus, manner, condition, or consequence.                                                                                  | May be true, incomplete, ambiguous, or false.                                           |
| Discoverability    | Empirical probability and process by which an agent can find an operation or relation under declared conditions.                                                       | Task-, observer-, context-, and time-relative.                                          |
| Believed action    | An action represented by an agent as available, with an associated confidence and expected consequence.                                                                | Latent; measured by elicitation or validated inference.                                 |



| Term          | Canonical definition                                                                                                                                 | Boundary                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Opportunity   | An umbrella analytical profile coordinating typed action-availability relations under declared system, agent, projection, task, and time conditions. | Not a new primitive and not synonymous with affordance or action set. |
| Reversibility | The degree to which consequences of an action can be undone, compensated, repaired, or safely explored under declared costs and constraints.         | May be exact, compensatory, partial, probabilistic, or unavailable.   |

## B. Opportunity Matrix

| Action                            | Domain-valid | Capable | Permitted | Enabled        | Reachable outcome  | Exposed      | Signaled  | Believed         | Selected     | Executed  | Successful    | Reversible |
|-----------------------------------|--------------|---------|-----------|----------------|--------------------|--------------|-----------|------------------|--------------|-----------|---------------|------------|
| GUI-hidden API export             | Yes          | Yes     | Yes       | Yes            | Yes                | No in GUI    | No in GUI | Expert-dependent | Context      | Context   | Context       | Usually    |
| Disabled submit                   | Yes          | Yes     | Yes       | No             | Possibly later     | Visible only | Yes       | Often            | No/attempted | Blocked   | No transition | N/A        |
| Unsupported conversational delete | No/absent    | No      | No        | No             | No                 | Apparent     | Yes       | Possible         | Possible     | Attempted | No            | N/A        |
| Role-prohibited export            | Yes          | Yes     | No        | Policy-blocked | No admissible path | May be       | May be    | Possible         | Possible     | Blocked   | No            | N/A        |

## C. Opportunity Contract

### Projection contract

#### Identity

- Declared target system, task, agent class, state scope, and comparison time.
- Projection identifier and modality.

#### System-relative action structure

- Domain-valid action vocabulary and transition semantics.
- Effective capability requirements.
- Permission and authority rules.
- Current enablement predicates.
- Reachable outcomes and path constraints.
- Failure modes and irreversible effects.

#### Projection-relative structure

- Exposed operations and access paths.
- Signifiers, labels, conventions, and feedback.
- Hidden prerequisites and external documentation.
- Ordering, composition, audit, and recovery support.

#### Observer assumptions

- Sensory and motor access.
- Language, conventions, training, and prior exposure.
- Available assistive technologies.
- Goal and risk context.

#### Epistemic variables

- Believed action distribution.
- Expected consequence distribution.
- Confidence and uncertainty.
- Perceived reversibility and cost.

#### Validity

- Intended use.
- Metrics and tolerances.
- Known omissions.
- Conditions under which comparisons fail.

## D. Opportunity Comparison Protocol

Representation comparison protocol

1. Declare the invariant target: operation semantics, task, initial state distribution, authority policy, and success criterion.
2. Verify system equivalence: confirm whether the projections share the same domain-valid, permitted, enabled, and reachable structures. Record any actual gating differences.
3. Enumerate projection exposure: identify every operation-access path, required sequence, modality, and prerequisite.
4. Code signaling: record perceptible cues, labels, conventions, state indications, feedback, and recovery cues.
5. Specify observers: measure relevant experience, language, sensory access, motor access, and training rather than using demographic proxies.
6. Measure belief: elicit availability judgments, expected consequences, confidence, and perceived reversibility before execution where possible.
7. Observe behavior: discovery, selection, execution, success, error, switching, and recovery.
8. Separate claims: classify every result as direct observation, derived system feature, or latent interpretation.
9. Compare typed relations: use set precision/recall, discovery curves, path cost, calibrated confidence, outcome distributions, and recovery metrics.
10. Test alternatives: control motor cost, latency, familiarity, content, task, implementation quality, and learning.
11. State scope: declare contexts and tolerances under which opportunity equivalence or inequivalence is claimed.

## E. Evidence Map

| Inference                      | Direct observation                                | Derived feature                 | Latent interpretation                | Alternatives                               | Falsifying evidence                                                |
|--------------------------------|---------------------------------------------------|---------------------------------|--------------------------------------|--------------------------------------------|--------------------------------------------------------------------|
| Action was exposed             | Executable path present and testable              | Projection-to-operation mapping | None required                        | Hidden prerequisite; inaccessible modality | Independent agents cannot execute via declared path                |
| Action was signaled            | Cue physically/linguistically present             | Cue-operation coding            | Cue interpreted as indication        | Cue decorative; ambiguous convention       | Removing cue does not change detection and agents deny association |
| Action was believed            | Availability judgment, prediction, confidence     | Belief-set estimate             | Internal availability representation | Demand effects; guessing; memory           | Judgments lack reliability or predictive validity                  |
| Agent hesitated                | Longer latency or pause                           | Duration relative to baseline   | Conflict or uncertainty              | Reading, motor delay, distraction          | Latency unchanged under uncertainty manipulation                   |
| Action was reversible          | Undo/repair path succeeds                         | Recovery graph and cost         | Perceived safety                     | Compensation incomplete; external effects  | Recovery fails or material effects persist                         |
| Projection changed opportunity | Differences in exposure/signaling/belief/recovery | Typed-set comparison            | Mediated access effect               | Motor cost, familiarity, latency           | Effect vanishes under controls                                     |

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